

# GIRISH KUMAR PATCHIKORU

Cloud Systems Engineer | AWS Infrastructure, Terraform & Kubernetes | Enterprise Data & AI/ML Platforms

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## PROFESSIONAL SUMMARY

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- **Cloud Systems Engineer with 9+ years of IT and cloud engineering experience**, including **6+ years hands-on with AWS**, provisioning, automating, securing, monitoring, and troubleshooting cloud infrastructure supporting **enterprise data, analytics, and AI/ML workloads** across healthcare, insurance, banking, and financial services.
- Deep hands-on experience across **AWS core services — EC2, S3, VPC, IAM, EKS, CloudWatch, CloudTrail, KMS, Lambda, Systems Manager, Parameter Store, Auto Scaling, ELB/ALB, Route 53, and CloudFront** — in regulated, audit-controlled enterprise environments.
- **5+ years building Infrastructure as Code with Terraform and AWS CloudFormation**, managing reusable modules, remote state, environment promotion, and drift detection for repeatable, version-controlled infrastructure provisioning.
- **4+ years operating Kubernetes and Amazon EKS in production**, covering cluster provisioning, node groups, Helm-based deployments, autoscaling, IRSA (IAM Roles for Service Accounts), networking, and containerized workload troubleshooting.
- Strong **AWS networking** background — VPC design, public/private subnets, route tables, NAT and Internet Gateways, security groups, NACLs, VPC endpoints, Transit Gateway, VPC peering, and hybrid connectivity.
- Extensive **IAM engineering** experience: roles, policies, trust relationships, permission boundaries, cross-account access, and **least-privilege** access models aligned to enterprise security and compliance requirements.
- Built and maintained **CI/CD pipelines** using **GitLab CI, GitHub Actions, Jenkins, and AWS CodePipeline/CodeBuild**, managing all infrastructure code through **Git/GitLab** repositories with peer review and automated validation gates.
- Implemented **cloud security, monitoring, logging, and automation** — CloudWatch metrics/alarms/log groups, CloudTrail audit logging, AWS Config, GuardDuty, encryption at rest and in transit via KMS, and secrets management.
- Proven troubleshooting record across **infrastructure, networking, deployments, IAM permissions, and performance issues**, with active participation in **incident response, root cause analysis (RCA), and continuous improvement**.
- Hands-on **automation and scripting with Python and Bash**, using AWS CLI, Boto3, and APIs to automate administrative tasks, patching, backups, capacity reporting, and **cloud cost optimization**.
- Practical experience supporting **enterprise data platforms and AI/ML workloads** in the cloud — Spark/Databricks, model serving, GPU/compute sizing, and production ML/GenAI services — bridging infrastructure engineering and data science delivery.
- **AWS Certified Solutions Architect – Associate** and **Google Professional Cloud Architect**; experienced working in **security-controlled and compliance-driven environments** with documented runbooks and architecture diagrams.

## TECHNICAL SKILLS

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- **AWS Cloud Services:** EC2, S3, VPC, IAM, EKS, ECS, ECR, CloudWatch, CloudTrail, AWS Config, KMS, Lambda, Auto Scaling, ELB/ALB/NLB, Route 53, CloudFront, RDS, EBS, EFS, SNS, SQS, Systems Manager (SSM), Parameter Store, Secrets Manager, Cost Explorer, AWS Organizations, AWS GovCloud
- **Infrastructure as Code (IaC):** Terraform, Terraform Modules, Terraform Remote State, AWS CloudFormation, Helm, Ansible, YAML, JSON, Infrastructure Automation, Drift Detection
- **Containers & Orchestration:** Kubernetes, Amazon EKS, Docker, Helm Charts, Kubernetes Networking, Pod Autoscaling (HPA/Cluster Autoscaler), Node Groups, IRSA, Container Registry (ECR), Containerized Workload Support
- **CI/CD & Version Control:** GitLab CI/CD, GitHub Actions, Jenkins, AWS CodePipeline, AWS CodeBuild, Git, GitLab, GitHub, Branching Strategy, Automated Testing, Blue/Green & Rolling Deployments, Release Management
- **Cloud Networking:** VPC Design, Subnets, Route Tables, Security Groups, NACLs, Internet Gateway, NAT Gateway, VPC Endpoints, VPC Peering, Transit Gateway, DNS/Route 53, Load Balancing, Hybrid Connectivity, TLS/SSL
- **Cloud Security & Compliance:** IAM Roles & Policies, Least-Privilege Access, Permission Boundaries, Cross-Account Access, Encryption at Rest & In Transit, KMS, Secrets Management, GuardDuty, AWS Config, Audit Logging, Security Hardening, Compliance Controls, Responsible AI Governance

- **Monitoring, Logging & Observability:** Amazon CloudWatch (Metrics, Alarms, Logs, Dashboards), CloudTrail, Log Aggregation, Alerting, Prometheus, Grafana, Application & Infrastructure Monitoring, Drift & Performance Monitoring, Incident Response, Root Cause Analysis (RCA)
- **Scripting & Automation:** Python, Bash/Shell Scripting, AWS CLI, Boto3, REST APIs, PowerShell (basic), Automated Remediation, Patch Automation, Backup Automation, Cost Optimization Automation
- **Operating Systems & Tooling:** Linux (RHEL, Amazon Linux, Ubuntu), Windows Server, Systemd, Cron, SSH, Networking Utilities, Jira, Confluence, ServiceNow
- **Enterprise Data & AI/ML Platforms:** Databricks, Apache Spark/PySpark, SQL, PL/SQL, MLflow, FastAPI, Model Serving Infrastructure, MLOps, LLMOps, Generative AI, RAG, LangChain, LangGraph, Vector Databases, Scikit-learn, XGBoost, Pandas, NumPy
- **Cloud Platforms:** Amazon Web Services (AWS), AWS GovCloud, Microsoft Azure, Google Cloud Platform (GCP)
- **Practices:** Capacity Planning, Cloud Cost Optimization, Scaling & High Availability, Disaster Recovery, Runbook & Technical Documentation, Architecture Diagrams, Agile/Scrum, On-Call Support, Continuous Improvement

## CERTIFICATIONS

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- **AWS Certified Solutions Architect – Associate** — Amazon Web Services | [Add issue date]
- **Google Professional Cloud Architect** — Google | Issued Feb 2026 · Expires Feb 2028
- **AWS Certified Cloud Practitioner** — Amazon Web Services | Issued Apr 2023
- **Oracle Cloud Infrastructure 2025 Certified Foundations Associate** — Oracle | Issued Aug 2025
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — Oracle | Issued Aug 2025
- **Microsoft Certified: Azure Fundamentals (AZ-900)** — Microsoft | Issued Dec 2022
- **Microsoft Certified: Azure AI Fundamentals** — Microsoft | Issued Dec 2022
- **Microsoft Certified: Azure Data Fundamentals** — Microsoft | Issued Dec 2022
- **Prompt Engineering & Programming with OpenAI** — Columbia+ | Issued Sep 2025

## PROFESSIONAL EXPERIENCE

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### Senior Cloud / AI-ML Engineer | UnitedHealth Group | Edina, Minnesota, US Dec 2025 – Present

- Provision, configure, and maintain **AWS cloud infrastructure** supporting an **enterprise data, analytics, and AI/ML platform**, including **EC2, S3, VPC, IAM, EKS, CloudWatch, and CloudTrail** across development, test, and production environments.
- Develop and maintain **Terraform** modules and **CloudFormation** templates automating provisioning of VPCs, subnets, security groups, IAM roles, EKS clusters, S3 buckets, and managed data services, eliminating manual configuration drift.
- Manage all infrastructure code through **Git/GitLab repositories**, enforcing branch protection, peer review, automated Terraform plan validation gates, and version-controlled promotion across environments.
- Support **containerized workloads on Kubernetes / Amazon EKS**, managing node groups, Helm releases, autoscaling policies, resource requests/limits, **IRSA** service-account roles, and rolling deployments for AI/ML inference services.
- Design and administer **AWS networking** — VPC subnets, route tables, NAT/Internet Gateways, VPC endpoints, security groups, and NACLs — enabling secure, private connectivity between data platform, analytics, and ML services.
- Implement **least-privilege IAM** roles, policies, permission boundaries, and cross-account trust relationships governing access to platform data, compute, and deployment pipelines.
- Configure **CloudWatch metrics, log groups, dashboards, and alarms** alongside **CloudTrail** audit logging to provide proactive alerting, operational visibility, and traceable audit evidence.
- Implement **secure cloud configurations** including **KMS encryption at rest and in transit**, TLS enforcement, secrets management via **Secrets Manager and SSM Parameter Store**, and hardened baseline configurations.
- Support and extend **CI/CD pipelines** (GitLab CI / GitHub Actions) automating infrastructure deployments, container image builds, security scanning, and controlled production releases.
- **Troubleshoot infrastructure, networking, deployment, IAM permission, and performance issues** across the platform, resolving pod scheduling failures, connectivity faults, throttling, and access-denied conditions.

- Automate administrative tasks using **Python (Boto3), Bash scripting, AWS CLI, and APIs**, covering patching, backups, resource tagging, log rotation, and environment cleanup.
- Support **cloud scaling, capacity planning, and cost optimization** — right-sizing EC2 and EKS node groups, tuning autoscaling, applying S3 lifecycle policies, and reporting spend through Cost Explorer.
- Maintain **technical documentation, runbooks, deployment procedures, and architecture diagrams**, and participate in **incident response, RCA, and continuous improvement** reviews.
- Architect and deploy production **Generative AI and RAG applications** using Python 3.12, LangChain, LangGraph, FastAPI, Databricks, Azure, and Google Cloud, running on containerized, cloud-native infrastructure.
- Implement **ML/LLM observability and evaluation frameworks** tracking response quality, groundedness, retrieval relevance, hallucination risk, data quality, and inference latency across deployed AI services.

**Environment:** AWS (EC2, S3, VPC, IAM, EKS, CloudWatch, CloudTrail, KMS, Lambda, SSM, Secrets Manager), Terraform, CloudFormation, Kubernetes, Docker, Helm, GitLab CI/CD, GitHub Actions, Git, Python 3.12, Bash, Boto3, AWS CLI, Linux, Databricks, Apache Spark/PySpark 3.5, FastAPI, REST APIs, LangChain 1.0, LangGraph 1.0, Azure, Google Cloud, Prometheus/Grafana.

#### **Cloud / AI-ML Engineer | Prudential Financial | Newark, NJ Jul 2024 – Nov 2025**

- Engineered and maintained **production AWS infrastructure** hosting underwriting, risk decisioning, and machine learning workloads, with primary ownership of **EC2, S3, VPC, IAM, EKS, and CloudWatch** resources.
- Authored reusable **Terraform** modules for VPCs, subnets, security groups, IAM roles/policies, S3 buckets, EKS clusters, and Auto Scaling groups, standardizing provisioning across multiple environments and accounts.
- Managed **remote Terraform state with S3 and DynamoDB locking**, implementing drift detection and controlled apply workflows to keep deployed infrastructure aligned with committed code.
- Deployed and operated **containerized model-serving microservices on Amazon EKS**, configuring Helm charts, horizontal pod autoscaling, ingress/ALB routing, liveness and readiness probes, and rolling upgrades.
- Built **CI/CD pipelines** automating infrastructure and application deployment — validation, security scanning, artifact promotion, and approval-gated production release across dev, test, and prod.
- Designed **least-privilege IAM roles and policies** for services, pipelines, and engineers, replacing broad permissions with scoped, auditable access aligned to internal security standards.
- Configured **VPC networking** — private/public subnets, route tables, NAT Gateways, VPC endpoints, and security groups — securing traffic between application, data, and inference tiers.
- Instrumented **CloudWatch alarms, dashboards, and centralized log groups**, monitoring latency, error rates, saturation, and service health; enabled **CloudTrail** for API-level audit logging.
- **Troubleshoot production infrastructure, deployment, permission, and performance incidents**, performing root cause analysis and driving remediation and preventive actions.
- Automated recurring operational tasks with **Python and Bash**, including environment refreshes, resource tagging compliance, snapshot/backup routines, and idle-resource cleanup for **cost optimization**.
- Built distributed data-processing pipelines using **Apache Spark/PySpark 3.5** and SQL, sizing and tuning cloud compute for high-volume insurance workloads.
- Managed **model lifecycle and experiment tracking with MLflow 2.14**, supporting versioning, reproducibility, and controlled promotion of production models.
- Established **model and data drift monitoring** and instrumented inference services for latency and failure detection, supporting operational health and RCA.
- Produced **governance, explainability, and compliance documentation** covering configurations, validation results, limitations, and observed production behavior for audit review.

**Environment:** AWS (EC2, S3, VPC, IAM, EKS, CloudWatch, CloudTrail, KMS, Auto Scaling, ALB, DynamoDB), Terraform, CloudFormation, Kubernetes, Docker, Helm, CI/CD, Git/GitLab, Python 3.11, Bash, Boto3, Linux, Apache Spark/PySpark 3.5, MLflow 2.14, FastAPI, REST APIs, Microservices, LangChain 0.3.

#### **Data Scientist (ML) / Cloud Engineering Support | TransUnion | Illinois, US Nov 2022 – Jun 2024**

- Deployed and supported analytical and machine learning workloads on **AWS**, working with **EC2, S3, IAM, and CloudWatch** to maintain reliable, secure, and cost-efficient execution environments.

- Contributed **Terraform** configurations provisioning compute, storage, and IAM resources for credit-risk and identity-resolution workloads, replacing manual console provisioning.
- Configured **S3 bucket policies, encryption, lifecycle rules, and IAM access controls** protecting large-scale consumer and credit datasets under strict regulatory requirements.
- Supported **CI/CD deployment pipelines** and **Git**-based version control for analytical code, models, and configuration, improving release consistency and traceability.
- Monitored job performance and infrastructure health via **CloudWatch metrics and logs**, troubleshooting failures, resource contention, and long-running workload performance issues.
- Developed **credit-risk machine learning models** using Python 3.10 and Scikit-learn based on credit behavior, payment history, utilization, tenure, and delinquency indicators.
- Optimized classification models through feature selection and hyperparameter tuning, evaluating precision, recall, F1-score, and ROC-AUC, and analyzing model stability across consumer segments.
- Built **identity-resolution and entity-matching workflows** using clustering and similarity techniques, generating confidence scores for record linking and deduplication.
- Supported **TruAudience and OneTru platform initiatives**, validating unified consumer attributes and model inputs across credit-risk and identity use cases.

**Environment:** AWS (EC2, S3, IAM, CloudWatch), Terraform, Git, CI/CD, Linux, Python 3.10, SQL, Scikit-learn 1.2, Machine Learning, Credit Risk Modeling, Identity Resolution, Feature Engineering, Model Validation, Statistical Analysis.

#### **Data Scientist / Cloud Platform Engineer | JPMorgan Chase & Co. (JPMC) | Texas, US Nov 2020 – Oct 2022**

- Partnered with Data Engineering on **legacy platform modernization**, migrating analytical datasets and workloads to **AWS** and **Databricks**, including EC2 compute, S3 storage layers, and IAM access design.
- Provisioned and configured **AWS infrastructure** for large-scale data processing — EC2 instances, S3 data lakes, IAM roles, security groups, and VPC subnet placement for regulated banking data.
- Implemented **infrastructure automation and scripted provisioning** using **Python, Bash, and AWS CLI**, reducing manual environment setup and standardizing cluster configuration.
- Configured **CloudWatch monitoring and logging** for Spark and Databricks jobs, alerting on failures, resource saturation, and cost anomalies.
- Applied **IAM least-privilege policies and S3 encryption** controls, aligning platform access with banking security and audit requirements.
- Developed **fraud and transaction-risk ML models** using Python 3.9 and Scikit-learn 1.0, detecting suspicious behavior across large banking datasets.
- Engineered transaction-level behavioral features with **Pandas, NumPy, and SQL**, including velocity, account activity, and rolling aggregates.
- Built **scalable feature-engineering workflows** using **Apache Spark 3.2** and Databricks over large structured banking datasets.
- Managed **experiment tracking and reproducibility with MLflow**, capturing parameters, metrics, and model versions across development cycles.

**Environment:** AWS (EC2, S3, IAM, VPC, CloudWatch), Databricks, Apache Spark 3.2, Python 3.9, Bash, AWS CLI, SQL, Pandas, NumPy, Scikit-learn 1.0, MLflow 1.23, Git, Linux, Tableau, Jupyter Notebook.

#### **Data Scientist / Systems Engineer | InfraSoftTech | Pune, India Apr 2017 – Sep 2020**

- Supported **Linux server environments and application deployments** for AML and banking compliance platforms, handling configuration, scheduled jobs, log management, and production troubleshooting.
- Automated deployment, monitoring, and batch operations using **Bash and Python scripting**, reducing manual intervention in recurring administrative tasks.
- Used **Git** for source control and participated in structured release, deployment, and production support processes.
- Developed **fuzzy and phonetic identity-matching workflows** in Python 3.6 for KYC screening against sanctions and Politically Exposed Person (PEP) watchlists.
- Applied **Levenshtein Distance and Jaro-Winkler** similarity algorithms to customer-name matching, improving verification across inconsistent records.

- Built anomaly-detection models with Scikit-learn using **Isolation Forest and One-Class SVM** to surface unusual transaction behavior beyond rule thresholds.
- Enhanced **Rule-Based Engine (RBE)** transaction monitoring with statistical outlier detection, extending AML coverage beyond deterministic rules.
- Developed fraud classification models using **XGBoost, Logistic Regression, and Random Forest**, applying **SMOTE** class balancing for rare-event detection.
- Built classical **NLP workflows** with NLTK, SpaCy, and TF-IDF supporting preprocessing, feature generation, and intent classification for banking chatbots.

**Environment:** *Linux, Bash, Python 3.6, Git, SQL, PL/SQL, Pandas, NumPy, Scikit-learn, XGBoost, Isolation Forest, One-Class SVM, SMOTE, K-Means, NLTK, SpaCy, TF-IDF, Jupyter Notebook, Confluence.*

## EDUCATION

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- **Bachelor of Technology (B.Tech), Computer Science** — KIET Group of Institutions, Uttar Pradesh, India